

6. Test Data Strategy – Performance & Load Testing



- When Teams often ask these questions:

"Do we have enough users in the test DB?"

"What kind of users do we need for a spike test?"

"Are we using mock data or prod-clone?"

- Without this, tests fail, show false positives, or waste hours debugging "data not found" errors.

1. Types of Data Required

Data Type	Purpose	Example
Users / Participants	To simulate login, search, transactions	10K active users, 5K inactive
Accounts / Plans	Investment scenarios	401K, 403B, 529, Roth IRA
Transactions	Throughput for APIs	500K insert/load
Roles / CSR Profiles	Access-level validation	Admin, Agent, Approver
Test API Tokens	Auth testing & security	JWTs, OAuth2

2. Test Data Strategy Overview

- < 1000 users
 - Reuse existing QA DB
 - Minimal role diversity
 - Use for sanity checks
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- 2K–10K users
 - Custom SQL or mock scripts
 - Aligned with API or screen flows
 - Ideal for pre-release validation
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- 50K+ users
 - DB clones, data masking from PROD
 - High diversity in accounts, roles, Txn types
 - Use when testing prod-like traffic

3. Tools & Scripts Used

Tool / Method	Purpose	Location
SQL Scripts	Load participants, txn data	/scripts/data_load.sql
Stored Procedures	Insert bulk CSR users	sp_insert_participants
CSV/Excel + JMeter CSV Data-Set	Parametrized inputs	/data/test_users.csv
BlazeMeter CSV Upload	Cloud script inputs	Linked in YAML configs
Mock APIs	Dummy data for low-risk APIs	/mockserver

4. DB Configuration & Environment Rules

Environment	DB Size Needed	Data Load Frequency	Notes
Stage	Medium (10K–50K)	Weekly refresh	Used for all soak/spike
Dev-QA	Small (2K–5K)	On-demand	Fast but less accurate
Perf-Clone	Large (100K+)	Monthly mask & clone	Mimics production

5. Access & Credentials

Type	Owner / Request Method
Stage DB Write Access	QA Lead / RT Ticket
Prod Clone Masking	DB Admin + Data Privacy
CSR Sample Users	Predefined in script repo
API Secrets	Jenkins Vault or DevEnv.yaml

6. Naming & Data Usage Standards

- Use naming conventions:
`perf\_user\_001`, `csr\_admin\_003`, `txn\_invest\_2025q2`
- Tag data with scenario in DB if possible (e.g., `test\_tag = 'perf\_baseline'`)
- Do ***not use real user data*** in shared envs
Use masked or dummy-generated only.
- Prefer ***SQL + Scripted ingestion*** over manual inserts.

7. Data Privacy & Compliance

- No real PII (name, SSN, DOB) in QA or Perf
- Masked values: `John D`, `1980-01-01`, `xxx-xx-4321`
- Use data privacy guidelines approved by InfoSec
- All clones from PROD must be logged and approved via RT

8. Templates & Reusable Artifacts

Attached Files & Templates
Sample 10K User Load SQL CSV for Investment Testing YAML using CSV Data Source CSR Login Account Pack Stored Proc Generator Script

9. Pro Tips & Gotchas

- Login test fails: CSR user not provisioned
- DB timeout: Trying to insert 50K at once — batch into 5K chunks
- Missing account types: Ensure diversity in load files
- API fails with 500: Token expired or invalid input data